

abrupt junction diode. A typical numerical computation [9] is reported in Fig. 10 where the symbols have a meaning similar to the doubler case and $\zeta = (3/2 R_c \omega_1)/m |q_0|$. The diagram of Fig. 10 is in qualitative agreement with the results shown in Fig. 6.

C. Transistor Circuits

From Figs. 2–4 it can be observed that the variation of phase shift versus power level in common-emitter transistor frequency multipliers has the opposite sign with respect to the other devices considered in this paper. The reason of this behaviour, although very difficult from a rigorous point of view, arises from the different parameters which influence the phase variation. In the case of varactor and step-recovery diodes the nonlinear process is due to a transition capacitance; in the transistor circuits an important parameter seems to be the diffusion capacitance of the input junction, which depends on the dc collector current.

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A Direct Time-Domain Measure of Frequency Stability: The Modified Allan Variance

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Abstract—A simple filter is shown to closely approximate the procedure used to compute the "Allan variance," an often used measure of the short term frequency stability of oscillators. This modified Allan variance is obtained by measuring the rms value of the instantaneous frequency waveform after filtering. Some advantages of the modified Allan variance include ease of implementation for short averaging times and better behavior when the instabilities include periodic components. The effect of the bandwidth of the measuring process on the minimum usable averaging time is also discussed.

I. INTRODUCTION

THE "ALLAN VARIANCE" for an oscillator is defined in terms of the fractional variation of the average frequency over intervals of length τ [1]–[3]. It is often estimated using frequency counters and digital computation. A direct method for estimating a modified Allan variance is described here.

At present, the Allan variance is estimated by computing the average of the instantaneous frequency of an oscillator over τ -seconds, and then computing the mean square value of the variation of this average frequency. This is obtained by taking the difference of the averaged frequency at times τ -seconds apart; and computing the mean square of these differences. Finally, this is divided by the average or "carrier" frequency (usually estimated by averaging the frequency over a time much longer than τ).

Following Barnes [3], the instantaneous fractional frequency is defined as

$$y(t) \equiv \frac{\dot{\psi}(t)}{2\pi f_s} \quad (1)$$

where $\dot{\psi}(t)$ is the derivative of the total instantaneous phase, and f_s is the average frequency of the signal from the oscillator

$$f_s = \lim_{T \rightarrow \infty} \left[\frac{\psi(T/2) - \psi(-T/2)}{2\pi T} \right]$$

Let $S_y(f)$ denote the one-sided spectral density on a per Hertz basis of the real function $y(t)$. The spectral density

of $y(t)$ can be computed from the spectral density of $\psi(t)$ as follows:

$$S_y(f) = \frac{1}{2\pi f_s} S_{\dot{\psi}}(f) = \frac{f^2}{f_s} S_{\psi}(f). \quad (2)$$

Barnes shows that the Allan variance, denoted by $\sigma_y^2(\tau)$, is related to the frequency domain frequency stability measure $S_y(f)$ by

$$\sigma_y^2(\tau) = 2 \int_0^\infty S_y(f) \frac{\sin^2(\pi\tau f)}{(\pi\tau f)^2} \left\{ 1 - \frac{\sin^2(2\pi\tau f)}{4 \sin^2(\pi\tau f)} \right\} df. \quad (3)$$

Equation (3) states that the Allan variance can be estimated by integrating the power spectrum of the instantaneous fractional frequency fluctuation after passing it through a filter with transfer function given by the expression under the integral in (3). The integration simply computes the total power, which can also be done by integrating in the time domain after filtering the signal $y(t)$ using the transfer function in (3).

Examination of this Allan variance filter shows that it consists of a $(\sin x)/x$ “low-pass” stage followed by a periodic transfer function which has a high-pass character at low frequencies. A plot of this transfer function is shown as the solid curve in Fig. 1. The parameter τ is often referred to as the “averaging time” and is of central importance since the frequency variation over some specified interval usually corresponds to some performance attribute of the system in which the oscillator is used. For example, in FMOP radar, the frequency variation from linear within the pulse determines the target resolution which can be achieved. Therefore, the frequency variations over time intervals corresponding to the radar’s resolution cell width are of major interest. Compare this to the pulse-Doppler radar, where coherent integration time (over several pulse intervals) determines the velocity resolution which can be achieved. In this case, frequency variation over time intervals corresponding to the reciprocal of the velocity resolution cell width is of major importance. The point is that, for some applications, averaging time intervals of nanoseconds or microseconds are important, for others variations over milliseconds or seconds is important, while for still others, variation over hours or even years is important. Short-term stability measurements generally deal with fractional frequency variation over averaging times of a few seconds or less.

II. APPROXIMATING THE ALLAN VARIANCE FILTER

The dashed curve in Fig. 1 shows the response of a filter which is easily implemented and which closely approximates the Allan variance filter transfer function. It consists of a simple C - R high-pass stage and a simple R - C low-pass stage with identical break points at $RC = \tau/2$. If the instantaneous frequency waveform is fed through such a filter and the rms value of the resulting voltage is measured, the result is an estimate of the square root of a modified Allan variance (after division by the nominal or

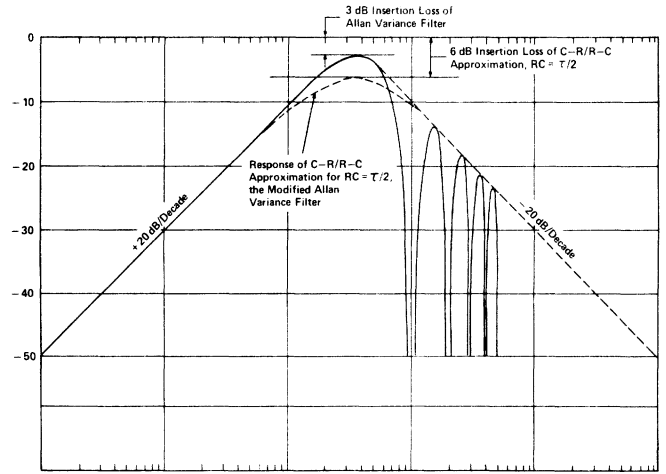


Fig. 1. Allan variance filter transfer function (τ = averaging time).

average frequency of the oscillator). The estimate is conservative since it includes all of the spectrum under the dotted portion of the curve in Fig. 1, without the deep notches in the Allan variance filter transfer function. However, it does have a 3 dB greater insertion loss at τ .

If the instantaneous total phase waveform is available (instead of the instantaneous frequency) the filter characteristic shown in Fig. 1 must be modified by a factor proportional to $1/f^2$ to account for the differentiation which occurs naturally in the discrimination process. To achieve the same result as obtained by filtering the instantaneous frequency waveform, the phase waveform must be filtered using a high pass slope characteristic of +40 dB/decade at low frequencies, and no low-pass characteristic to diminish the response at the higher frequencies. The problem of approximating the filter for processing the phase data has been addressed by Rutman [4]. However, if the instantaneous frequency waveform is available, these filters are not needed. Consider the case when $S_y(f)$ is constant (i.e., the frequency instability is white noise). Then the ideal Allan variance estimate from (3) would be

$$\sigma^2(\tau) = 2 \int_0^\infty \frac{\sin^2(\pi\tau f)}{(\pi\tau f)^2} df - 2 \int_0^\infty \frac{\sin^2(2\pi\tau f)}{(2\pi\tau f)^2} df \quad (4)$$

$$= \frac{1}{2\tau}. \quad (5)$$

Using simple R - C high-pass and low-pass filters would yield a modified Allan variance (for white noise) of

$$\hat{\sigma}^2(\tau) = 2 \int_0^\infty \left[\frac{1}{1 + \frac{1}{(\pi f \tau)^2}} \right] \left[\frac{1}{1 + (\pi f \tau)^2} \right] df \quad (6)$$

$$= 2 \int_0^\infty \frac{(\pi f \tau)^2}{[1 + (\pi f \tau)^2]^2} df. \quad (7)$$

Let

$$u = 1 + (\pi f \tau)^2 \text{ and } du = 2(\pi\tau)(\pi\tau f) df \quad (8)$$

$$\hat{\sigma}^2(\tau) = \frac{2}{\pi\tau} \int_1^\infty \frac{(u-1)^{1/2}}{2u^2} du \quad (9)$$

$$= \frac{2}{\pi\tau} \left[\frac{(u-1)^{3/2}}{u} - (u-1)^{1/2} + \tan^{-1}(u-1)^{1/2} \right]_1^\infty \quad (10)$$

$$= 1/\tau \text{ (for } RC = \tau/2). \quad (11)$$

Therefore, when the instantaneous frequency waveform is white noise, the simple cascade C - R high-pass and R - C low-pass gives exactly twice the mean square output as the Allan variance filter. Although the RC approximation has a 3 dB larger insertion loss, this only partially compensates for the lack of attenuation "notches" at frequencies $f = N/\tau$; $N = 1, 2, 3, \dots$. Of course, if the spectrum of the frequency fluctuation is not white, the approximate filter may provide an estimate for the Allan variance which is greater or smaller than the actual Allan variance. By simply incorporating an attenuation of 0.5, the modified Allan variance as measured at the output of the RC filter will be equal to the unmodified Allan variance when the frequency disturbance is white noise.

The notches in the bandpass characteristics of the Allan variance filter are an obvious consequence of the definition: frequency variations which are periodic such that the period is a submultiple of the averaging time τ contribute nothing to the Allan variance. For instance, if $\tau = 1/60$ second, then any 60 Hz (or multiples of 60 Hz) variations in frequency will not affect the Allan variance for $\tau = 1/60$. Also large changes in the Allan variance for small changes in averaging time are possible if periodic frequency variations are present. For most applications, this appears to be a disadvantage to using the Allan variance. Certainly it should be carefully measured over a range of averaging times to avoid problems with periodic frequency fluctuations. Indeed, the modified Allan variance computed by the simple C - R / R - C filter combination appears to be the more desirable measure of oscillator stability, with the presently defined Allan variance being a second choice which is easy to calculate for sampling measurement systems (e.g., frequency counters). Certainly, for short averaging times, the modified Allan variance suggested here has many advantages.

III. LIMITATIONS DUE TO BANDLIMITED MEASUREMENT SYSTEMS

In any real measurement system, the instantaneous frequency (or phase) is determined over a limited bandwidth. Thus, the Allan variance cannot be estimated for arbitrarily short averaging times (τ). In fact, for bandlimited signals, the variation in a time τ is bounded. Papoulis [5] has shown that a random ergodic signal, bandlimited to f_b having an average power P has a mean square variance over time τ which is bounded by

$$\sigma_y^2(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} (\bar{y}(t) - \bar{y}(t + \tau))^2 dt < (2\pi f_b \tau)^2 P \quad (12)$$

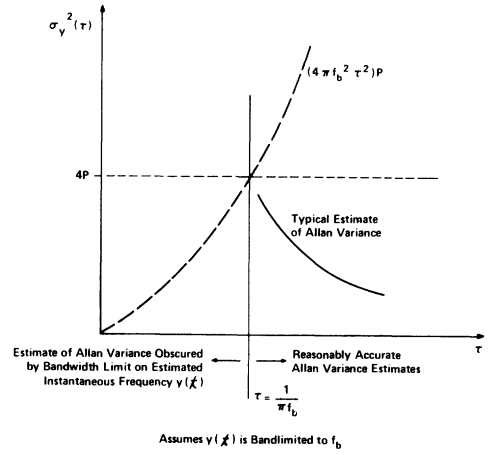


Fig. 2. Bounds on Allan variance estimate for a bandlimited measurement system.

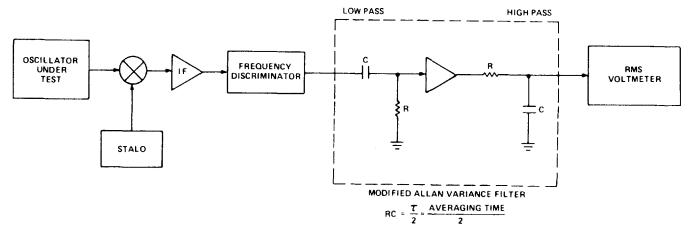


Fig. 3. Instrumentation for modified Allan variance measurement.

where $\bar{y}(t)$ is the instantaneous frequency averaged over time τ

$$\bar{y}(t) = \int_{-\infty}^{\infty} y(t - \tau) p(\tau) d\tau$$

$p(t)$ is a unit pulse extending from $t = -\tau/2$ to $t = +\tau/2$, and

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{+T/2} \bar{y}^2(t) dt = \text{average power of the } \tau\text{-averaged fractional frequency.}$$

Expanding the integrand in (12) gives another bound on $\sigma_y^2(\tau)$.

$$\sigma_y^2(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \left[\int_{-T/2}^{+T/2} (\bar{y}^2(t) - 2\bar{y}(t)\bar{y}(t + \tau) + \bar{y}^2(t + \tau)) dt \right]. \quad (13)$$

The center term in the integrand of (13) is the correlation $R_{yy}(\tau)$, which can be no larger than $R_{yy}(0) = P$. Thus

$$\sigma_y^2(\tau) \leq 4P. \quad (14)$$

Assuming that the estimate of $y(t)$ is subject to an upper frequency cutoff f_b , the Allan variance estimated by such a measurement system is subject to the bounds shown in Fig. 2, which graphically illustrates (12) and (14). Clearly, reasonably accurate Allan variance estimates can be made

only for

$$\tau > 1/\pi f_b. \quad (15)$$

IV. SUMMARY

There are advantages to slightly modifying the Allan variance definition to eliminate the effects of periodic sampling. When this is done, the resulting modified Allan variance is easy to estimate using the output of a frequency discriminator as shown in Fig. 3. For short averaging times (τ) of less than one second, the modified Allan variance is a useful measure of oscillator quality which is easier to estimate than the presently defined Allan variance.

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Thermistor Multivibrator as the Temperature-to-Frequency Converter and as a Bridge for Temperature Measurement

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Abstract—The symmetrical multivibrator, one timing resistor of which is replaced by a network consisting of a thermistor and passive resistors, is analyzed. The circuit represents at the same time a linear temperature-to-frequency (T - f) converter and a multivibrator bridge (MB), both having the same linearity range at any thermistor network configuration. When the balancing state is reached at an inflection point, the sensitivities of the converter and the bridge are the same. The MB has higher sensitivity than the Wheatstone bridge (WB) and can be balanced independently of the inflection point in the wide range. The converter and bridge performances with practical thermistor transducers are analyzed theoretically and numerically.

INTRODUCTION

DIRECT THERMISTOR temperature-to-frequency (T - f) converters based on relaxation oscillators have been successfully applied in small telemetering systems and instrumentation for measurement and integration of temperature [1]–[5]. Direct converters, though simple in construction, have the stability and sensitivity of characteristics similar to those of more complex indirect T - f converters based on thermistor bridges and voltage-to-frequency converters. This paper describes a direct T - f converter having the form of a symmetrical thermistor multivibrator which can be also applied as a bridge circuit for temperature measurement by means of the dc voltmeter. In [5], it was shown that the classical astable multivibrator, with the thermistor as one timing resistance, could be used as the linear T - f converter. Paper [6] proved

that when the same circuit is applied as the bridge, it has the same linearity range as that of T - f converter. Both the converter and the bridge with the given thermistor and the defined temperature range, in this basic configuration, have the fixed sensitivity which cannot be regulated. At the constant supply voltage, the thermistor dissipation is also uncontrollable, which may cause a serious measuring error due to the thermistor selfheating. The balancing state of the multivibrator bridge (MB) in the basic configuration is obtained at the temperature always higher than the inflection temperature [6]. This represents a disadvantage in comparison with the thermistor Wheatstone bridge (WB), the balancing condition of which is independent of the inflection point [7].

The applied circuit here is the classical multivibrator. But, the results obtained can be used also with other symmetrical multivibrators having two complementary outputs and two timing resistors, as for instance the multivibrator with two current generators [8], or the multivibrator with two regenerative switches [9], [10].

CLASSICAL THERMISTOR MULTIVIBRATOR AS THE T - f CONVERTER AND BRIDGE CIRCUIT

Fig. 1 displays the multivibrator, the frequency and the output voltage of which are measured by means of high-impedance instruments. In the general case, the timing resistor $R_e(T)$ can be substituted by the network composed of a thermistor $R(T) = R(T_1) \exp B(1/T - 1/T_1)$ and of fixed resistors. If the thermistor dissipation is negligible, the equivalent resistance of the network with an arbitrary configuration, can be expressed in the form

$$R_e(T) = \frac{A_1 R(T) + A_2}{A_3 R(T) + A_4} \quad (1)$$

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